A MEROMORPHIC FUNCTION VANISHING ON A PRINCIPAL OPEN IS ANNIHILATED BY A POWER OF THE DEFINING SECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 6241-6246 and proof span 6248-6258 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof uses the author's notation for the sheaf of meromorphic functions and for the set of nonzerodivisors on an affine open; both are preserved.

Allowed standard prerequisites (editorial).

- The sheaf of meromorphic functions is described locally on affine opens as quotients of a section by a nonzerodivisor, and the structure sheaf embeds into it. Used at proof lines 6248-6252 and again at 6255-6257. Allows writing the meromorphic function locally as a quotient and lets a section of the structure sheaf be viewed inside the meromorphic sheaf.
- On an affine scheme, killing a section after inverting an element is equivalent to annihilating it by a power of that element. Used at proof lines 6256-6258. Converts vanishing of the image in the localization into the annihilation statement.
- Quasi-compactness reduces a global statement to finitely many affine pieces. Used at proof lines 6248-6252. Lets the proof replace the scheme by one affine open on which the meromorphic function has the described form.

Lemma 0.1. *Let X be a quasi-compact scheme. Let $h \in \Gamma(X, \mathcal{O}_X)$ and $f \in \Gamma(X, \mathcal{K}_X)$ such that f restricts to zero on X_h. Then $h^n f = 0$ for some $n \gg 0$.*

Proof. We can find a covering of X by affine opens U such that $f|_U = s^{-1}a$ with $a \in \mathcal{O}_X(U)$ and $s \in \mathcal{S}(U)$. Since X is quasi-compact we can cover it by finitely many affine opens of this form. Thus it suffices to prove the lemma when $X = \mathrm{Spec}(A)$ and $f = s^{-1}a$. Note that $s \in A$ is a nonzerodivisor hence it suffices to prove the result when $f = a$. The condition $f|_{X_h} = 0$ implies that a maps to zero in $A_h = \mathcal{O}_X(X_h)$ as $\mathcal{O}_X \subset \mathcal{K}_X$. Thus $h^n a = 0$ for some $n > 0$ as desired. □

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